

**RAMCO INSTITUTE OF TECHNOLOGY**  
**Department of Electronics and Communication Engineering**  
**Academic Year: 2019- 2020 (Odd Semester)**

**Innovative Practices Description**

**UNIT- I (8085 PROCESSOR)**

**Degree, Semester& Branch:** V Semester B.E. EEE

**Course Code & Title:** EE8551 Microprocessors and Microcontrollers

**Name of the Faculty member:** Mr.A.Rameshbabu

**Name of the Topic:** Hardware Architecture of 8085 microprocessor

**Name of the Innovative Practice:** Animation Video

**Date & Duration:** 03.07.2019 & 10 minutes

**ICT Tool Used:** Desktop, projector

**Description:**

Animation is a brilliant and innovative new way to encourage students to communicate stories, ideas and concepts in a creative and original way. It can be particularly useful as a tool to encourage the creativity of students. Animated videos are a one stop solution that can grab the attention of the students. With its entertaining visuals, it can clearly comprehend the concept. Since it is an analogy to cartoons, students will get hooked to it. By using animation videos, students learn analytical thinking as to how to present a scenario.

**Goals (Learning Outcomes):**

- The students will be able to easily identify the blocks of Hardware Architecture of 8085 microprocessor.
- The students will be able to explain the working function each block in the Hardware Architecture of 8085 microprocessor.

**Use of appropriate methods:**

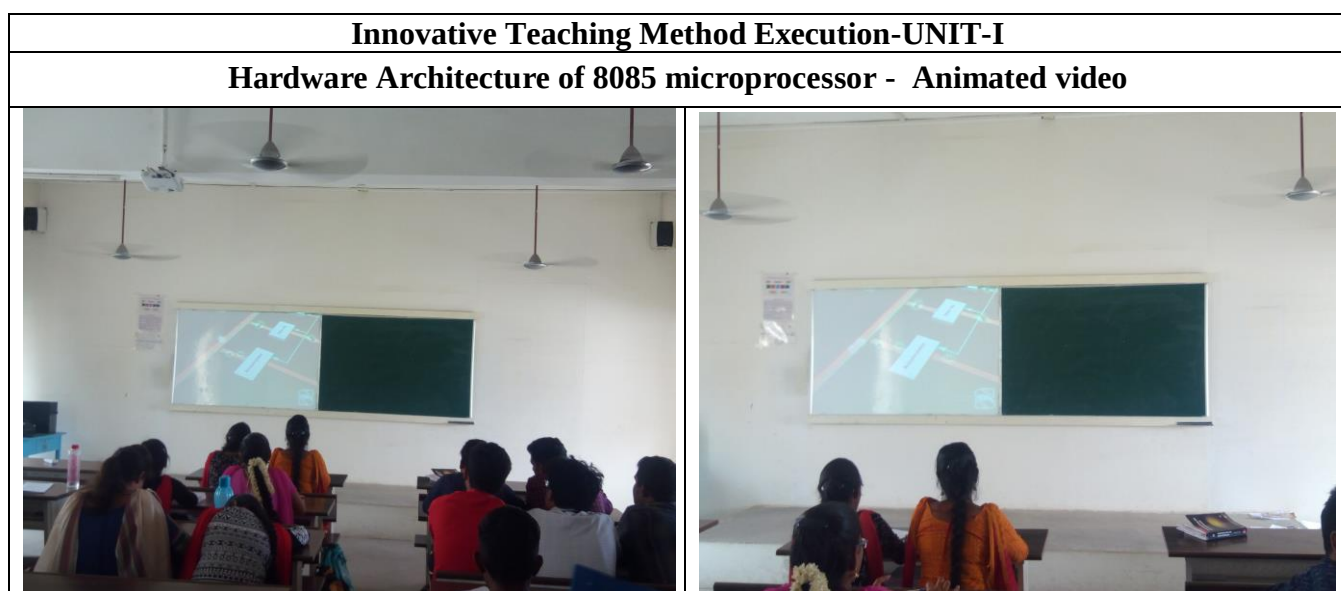
**Justification for choosing the topic using animation video activity:**

Stepper motor control concept is important topic in Microprocessor. Animated video helps to show how the internal blocks of Hardware Architecture of 8085 microprocessor works and their applications.

**Effective presentation**

**Details of Implementation:**

The suitable animated video for Hardware Architecture of 8085 microprocessor was displayed to the students and explained about the different types of blocks in the Hardware Architecture of 8085 microprocessor and their working functions.



### **Significance of Results:**

#### **Assessment of Effectiveness/Success of the Activity:**

- The success of the activity was evaluated by asking the same question in Internal Assessment test III – Around 85% of students answered correctly.

#### **Reflective Critique:**

- **Benefits:**

The students are more interested to watch animated video, apart from theory concept. Animation videos for 8085 processor help the students to know how the 8085 processor works. It helps to differentiate the different types of internal blocks in the Hardware Architecture of 8085 microprocessor.

- **Challenges:**

It takes some time to choose relevant videos for the Hardware Architecture of 8085 microprocessor topic.

#### **Changes required for future (if required):**

| Course Outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO1            | 1   | 1   | -   | -   | 1   | -   | -   | 1   | -   | -    | -    | 1    |

*CO1: The students will be able to develop applications using 8051 microcontroller.*

#### **Reference:**

1. Soumitra Kumar Mandal, Microprocessor & Microcontroller Architecture, Programming & Interfacing using 8085,8086,8051,McGraw Hill Edu,2013.
2. <https://www.youtube.com/watch?v=I78iyzXQrP4>

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**Innovative Practices Description**

**Unit – II (PROGRAMMING OF 8085 PROCESSOR)**

**Degree, Semester& Branch:** V Semester B.E. EEE

**Course Code & Title:** EE8551 Microprocessors and Microcontrollers

**Name of the Faculty member:** Mr.A.Rameshbabu

**Name of the Topic:** Control Instructions

**Name of the Innovative Practice:** Mind maps

**Date & Time:** 31.07.2019 & 15 Minutes

**ICT Tool Used:** Black Board

**Description:**

**Mind Map Technique:**

A mind map is an easy way to brainstorm thoughts organically without worrying about order and structure. It allows the student to visually structure the ideas which is used to help with analysis and recall.

A mind map is a diagram for representing tasks, words, concepts, or items linked to and arranged around a subject using a non-linear graphical layout that allows the student to build an intuitive framework around a central concept.

**Goals (Learning Outcomes):**

1. Students will be able to understand about Control Instruction functions.
2. Students will be able to explain about different types of Control Instructions.

**Use of appropriate methods:**

**Justification for choosing the topic using mind map activity:**

Mind map activity gives the graphical representation of Control Instructions. It differentiates 8085 Control Instructions clearly which will be helpful for understanding the concepts of 8085 Control Instruction functions to the students.

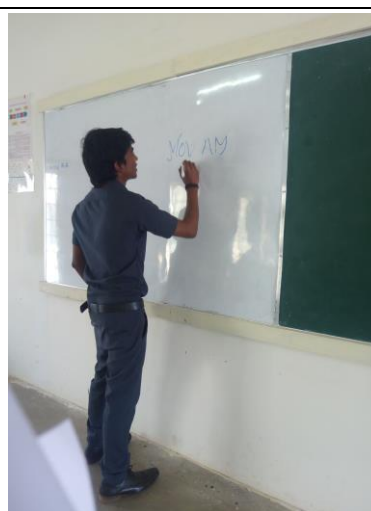
**Effective presentation**

**Details of the Implementation:**

The students confused about the different types of Control Instructions which was explained in the class room. After that, I have chosen Mind map as an Innovative practice method for better understanding about 8085 Control Instructions. The Students eagerly participated in that activity and then they acquired the concept of Control Instruction functions.

## Innovative Teaching Method Execution

Control Instructions :- Mind maps



### Significance of Results:

#### **Assessment of Effectiveness/Success of the Activity:**

- Every students got equal opportunity to come forward to take part in this activity
- The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correctly.

### Reflective Critique:

#### • **Benefits:**

Making a mind map is an excellent way for students to be able to sort through your thoughts and ideas. This activity allows the students to quickly generate creative and even unique ideas in less time. With the help of this activity the students can able to easily differentiate different types of Control Instructions.

#### • **Challenges:**

Initially, I have planned the activity for 10 minutes but some students not aware of this activity. So, that I have explained the activity and then some of the students hesitated for doing that. I encouraged the students which it takes another 5 minutes for completing the activity.

**Changes required for future (if required):** Planned to increase the Time Duration

| Course Outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO2            | 1   | 1   | 2   | -   | 1   | -   | -   | 1   | -   | 1    | -    | 1    |

*CO2: The students will be able to Design and implement programs on 8085 microprocessor.*

#### **References:**

- 1) Soumitra Kumar Mandal, Microprocessor & Microcontroller Architecture, Programming & Interfacing using 8085,8086,8051,McGraw Hill Edu,2013.
- 2) <https://www.qualitymatters.org/qa-resources/resource-center/articles-resources/mind-maps>
- 3) <https://youtu.be/u5Y4pIsXTV0>

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**Unit – III (8051 MICRO CONTROLLER)**

**Degree, Semester& Branch:** V Semester B.E. EEE

**Course Code & Title:** EE8551 Microprocessors and Microcontrollers

**Name of the Faculty member:** Mr.A.Rameshbabu

**Name of the Topic:** Timing Diagram

**Name of the Innovative Practice:** One minute paper:

**Date & Duration:** 27.08.2019 & 1 minute

**ICT Tool Used:-**

**Description:**

**One minute paper:**

The Minute Paper is a very commonly used classroom assessment technique. It really does take about a minute and, while usually used at the end of class, it can be used at the end of any topic discussion. Its major advantage is that it provides rapid feedback on whether the professor's main idea and what the students perceived as the main idea are the same. Additionally, by asking students to add a question at the end, this assessment becomes an integrative task. Students must first organize their thinking to rank the major points and then decide upon a significant question. Sometimes, instead of asking for the main point, a professor may wish to probe for the most disturbing or most surprising item. It is thus a very adaptable tool.

**Goals (Learning Outcomes):**

- Students will able to understand the concepts of Timing Diagram.
- Students will able to draw Timing Diagrams for different types of instructions.

**Use of appropriate methods:**

**Justification for choosing the One minute paper Activity:**

This Activity will take only one minute time duration. It is easy to recall all point in that particular topic. Timing Diagram topic is an important topic in unit 3 that was the reason behind to choose this topic for One Minute Paper activity.

**Time duration & Challenges:**

Initially, I have planned this activity for only one minute but students were taken one minute to become ready for this activity. So, that I have extent the activity for Two minutes. Next time I want to give prior information about this activity to my students to reduce the time duration.

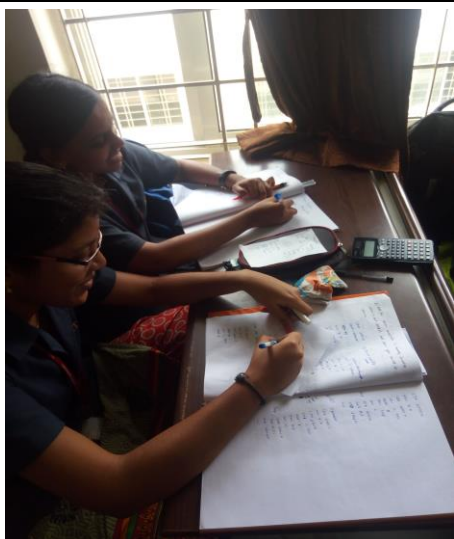
**Effective presentation**

**Details of the Implementation:**

The students confused to the Timing Diagram of 8051 instructions which was explained in the class room. After that, I have chosen One Minute Paper as an Innovative practice method for better understanding about Timing Diagrams. The Students eagerly participated in that activity and then they acquired the concept of Timing Diagrams.

### Innovative Teaching Method Execution

Timing Diagram :- One minute paper



#### Significance of Results:

##### **Assessment of Effectiveness/Success of the Activity:**

- After conducting this activity I found that many students understand this topic.
- The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correctly.

##### Reflective Critique:

- **Benefits:**  
Minute papers are an effective way of involving all students in class simultaneously. It ensures equal participation of each and every class member, including anyone who may be too shy or fearful to participate orally. It can promote class attendance and attentiveness. It can improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.
- **Challenges:**  
After teaching this particular topic students asked some time to prepare. So, I gave few minutes to prepare which takes more time to implement this activity than planned.

**Changes required for future (if required):** Nil

| Course Outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO3            | 1   | 1   | -   | -   | 1   | -   | -   | 1   | -   | 1    | -    | -    |

*CO3: The students will be able to elaborate the operation of 8051 microcontroller architecture.*

#### **References:**

- 1) R.S. Gaonkar, 'Microprocessor Architecture Programming and Application', with 8085, Wiley Eastern Ltd., New Delhi, 2013.
- 2) <https://oncourseworkshop.com/self-awareness/one-minute-paper/>
- 3) [https://www.youtube.com/watch?v=R\\_nf0J3i1w](https://www.youtube.com/watch?v=R_nf0J3i1w)

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**UNIT IV (PERIPHERAL INTERFACING)**

**Degree, Semester& Branch:** V Semester B.E. EEE

**Course Code & Title:** EE8551 Microprocessors and Microcontrollers

**Name of the Faculty member:** Mr.A.Rameshbabu

**Name of the Topic:** A/D and D/A Converters & Interfacing With 8085

**Name of the Innovative Practice:** Brainstorming

**Date & Time:** 23.09.19 & 30 Minutes

**ICT Tool Used:** - Desktop computer, projector

**Description:**

Brainstorming is a group creativity technique by which efforts are made to find a conclusion for a specific problem by gathering a list of ideas spontaneously contributed by its members. All the ideas are noted down without criticism and after the brainstorming session the ideas are evaluated.

**Goals (Learning Outcomes):**

- The students will be able to develop their creative thinking and interfacing skills.
- The students will be able to know about modern ADC & DAC used in industries.

**Use of appropriate methods:**

**Justification for choosing the topic using Brainstorming activity:**

A/D and D/A Converters & Interfacing With 8085 are important topics in unit 4 so I choose Brain storming activity to explain about this topic. The students already learned about this concept in previous chapter. So I asked them to prepare and share their ideas.

**Effective presentation**

**Details of Implementation:**

The process begins with a well-defined topic. Students are divided into groups. Each participant in a group creates an individual brainstorm around the topic. Each group members need to present their views on stage. All the ideas are then merged into one large idea map. During this consolidation phase the participants may discover a common understanding of the issues as they share the meanings behind their ideas. As the sharing takes place, new ideas may arise by the association. Those ideas are added to the map as well. Then ideas are generated on both individual and group levels.

## Innovative Teaching Method Execution

### **A/D and D/A Converters & Interfacing With 8085 - Brainstorming**



#### **Significance of Results:**

##### **Assessment of Effectiveness/Success of the Activity:**

- The success of the activity was evaluated by asking the same question in Internal Assessment test III – Around 80% of students answered correctly.
- Slow learners came forward to explain their ideas.

#### **Reflective Critique:**

##### **• Benefits:**

Brainstorming builds involvement, commitment, loyalty, and enthusiasm. Participating in the sessions stimulates and unlocks people's creative talents. Brainstorming also builds self-esteem because people are being asked for their participation and their ideas.

##### **• Challenges:**

For each student i allotted for 5 minutes to present their ideas on the stage. But few students take more than the time allotted.

**Changes required for future (if required):** Plan to deliver detailed explanation about this activity

| Course Outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO4            | 1   | 1   | -   | -   | 2   | -   | -   | 1   | -   | 1    | -    | 1    |

*CO4: The students will be able to describe the commonly used peripheral interfacing.*

#### **References:**

1. Soumitra Kumar Mandal, Microprocessor & Microcontroller Architecture, Programming & Interfacing using 8085,8086,8051,McGraw Hill Edu,2013.
2. <https://www.youtube.com/watch?v=YXZamW4-Ysk>
3. <https://technologyadvice.com/blog/marketing/brainstorming-activities-inspire-content-marketing-team/>



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**Innovative Practices Description**

**UNIT-V (MICRO CONTROLLER PROGRAMMING & APPLICATIONS)**

**Degree, Semester& Branch:** V Semester B.E. EEE

**Course Code & Title:** EE8551 Microprocessors and Microcontrollers

**Name of the Faculty member:** Mr.A.Rameshbabu

**Name of the Topic:** Stepper motor control

**Name of the Innovative Practice:** Animation Video

**Date & Duration:** 30.09.2019 & 10 minutes

**ICT Tool Used:** Desktop, projector

**Description:**

Animation is a brilliant and innovative new way to encourage students to communicate stories, ideas and concepts in a creative and original way. It can be particularly useful as a tool to encourage the creativity of students. Animated videos are a one stop solution that can grab the attention of the students. With its entertaining visuals, it can clearly comprehend the concept. Since it is an analogy to cartoons, students will get hooked to it. By using animation videos, students learn analytical thinking as to how to present a scenario.

**Goals (Learning Outcomes):**

- The students will be able to easily identify the different types of Stepper motor control functions.
- The students will be able to explain the application of Stepper motor control.

**Use of appropriate methods:**

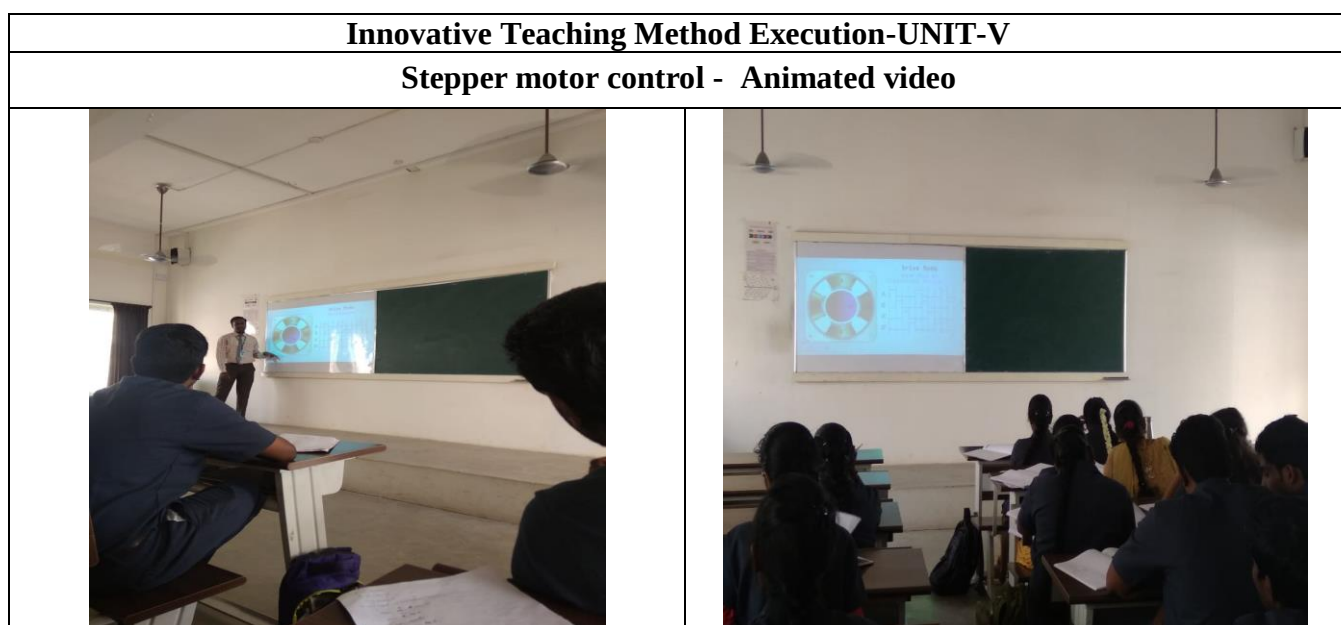
**Justification for choosing the topic using animation video activity:**

Stepper motor control concept is important topic in Microprocessor. Animated video helps to show how the Stepper motor control works and its real time applications.

**Effective presentation**

**Details of Implementation:**

The suitable animated video for Stepper motor control was displayed to the students and explained about the different types of Stepper motor control working functions and their real time applications.



### **Significance of Results:**

#### **Assessment of Effectiveness/Success of the Activity:**

- The success of the activity was evaluated by asking the same question in Internal Assessment test III – Around 85% of students answered correctly.

#### **Reflective Critique:**

##### **• Benefits:**

The students are more interested to watch animated video, apart from theory concept. Animation videos for Stepper motor help the students to know how the Stepper motor works. It helps to differentiate the different types of Stepper motor control functions.

##### **• Challenges:**

It takes some time to choose relevant videos for the Stepper motor control topic.

#### **Changes required for future (if required):**

| Course Outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO5            | 1   | 1   | -   | -   | 1   | -   | -   | 1   | -   | -    | -    | 1    |

*CO5: The students will be able to develop applications using 8051 microcontroller.*

#### **Reference:**

1. Soumitra Kumar Mandal, Microprocessor & Microcontroller Architecture, Programming & Interfacing using 8085, 8086, 8051, McGraw Hill Edu, 2013.
2. <https://www.youtube.com/watch?v=eyqwLiowZiU>